



TITEL

2. INTRODUCTION

MIKROELEKTRONIKA

Capacitors are common components of electronic circuits, used almost as frequently as resistors. The basic difference between the two is the fact that capacitor resistance (called reactance) depends on the frequency of the signal passing through the item. The symbol for reactance is X_c and it can be calculated using the following formula:

$$X_c = \frac{1}{2\pi fC}$$

f representing the frequency in Hz and C representing the capacitance in Farads.

For example, 5nF-capacitor's reactance at $f=125\text{kHz}$ equals:

$$X_c = \frac{1}{2 \cdot 3,14 \cdot 125000 \cdot 5 \cdot 10^{-9}} = 255 \Omega,$$

while, at $f=1.25\text{MHz}$, it equals:

$$X_c = \frac{1}{2 \cdot 3,14 \cdot 1250000 \cdot 5 \cdot 10^{-9}} = 25,5 \Omega.$$

A capacitor has an infinitely high reactance for direct current, because $f=0$.

Capacitors are used in circuits for many different purposes. They are common components of filters, oscillators, power supplies, amplifiers, etc.

The basic characteristic of a capacitor is its capacity – the higher the capacity, the higher is the amount of electricity it can hold. Capacity is measured in Farads (F). As one Farad represents fairly high capacity, smaller values such as microfarad (μF), nanofarad (nF) and picofarad (pF) are commonly used. As a reminder, relations between units are:

$$1\text{F}=10^6\mu\text{F}=10^9\text{nF}=10^{12}\text{pF},$$

that is $1\mu\text{F}=1000\text{nF}$ and $1\text{nF}=1000\text{pF}$. It is essential to remember this notation, as same values may be marked differently in some circuits. For example, 1500pF is the same as 1.5nF , 100nF is $0.1\mu\text{F}$.

A simpler notation system is used as with resistors. If the mark on the capacitor is 120 the value is 120pF , 1n2 stands for 1.2nF , n22 stands for 0.22nF , while $.1\mu$ (or $.1\text{u}$) stands for $0.1\mu\text{F}$.

Capacitors come in various shapes and sizes, depending on their capacity, working voltage, type of insulation, temperature coefficient and other factors. All capacitors can be divided in two groups: those with changeable capacity values and those with fixed capacity values. These will be covered in the following chapters.



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